

Signed circulants at the Ramanujan bound

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Abstract

For the circulant graph $C_n(1, 2)$ with $n \geq 10$ even, the $\mathbb{F}_2$ system requiring every quadrilateral to be unbalanced is consistent and its solutions form exactly four switching classes. We show that the class containing the signing which is $+1$ on step-1 edges and $(-1)^i$ on step-2 edges has spectrum $\{\pm 2\sqrt{\cos^2 \theta_k + \cos^2 2\theta_k}\}$ and spectral radius exactly $2\sqrt{2}$, well below the Kesten bound $2\sqrt{3}$; that the quadrilateral system is equivalent to alternating triangle fluxes, so that the four classes are coordinatized by (τ_0, α) and the spectral radius depends only on the Hamilton-cycle holonomy α ; and that the two twisted classes attain $\rho_-(n) = 2\sqrt{\cos^2(\pi/n) + \cos^2(2\pi/n)} < 2\sqrt{2}$. Exhaustive enumeration of all 2^{n+1} switching classes for $n \in \{8, 10, 12, 14, 16, 18\}$ shows that $\rho_-(n)$ is the global minimum in every case, and we conjecture this for all even n ; the lower bound is a flux-minimization statement in the sense of Lieb's flux-phase theorem. For odd n the quadrilateral system is inconsistent.

1 Setup

Let $G = (V, E)$ be a connected graph with n vertices and m edges and let $\sigma \in \{\pm 1\}^E$ be a signing of its adjacency matrix, encoded by $s \in \mathbb{F}_2^E$ with $\sigma_e = (-1)^{s_e}$. The spectrum of the signed adjacency matrix A_σ is invariant under switching, and the switching class of σ is determined by the signs $(-1)^{s \cdot \chi_C}$ of cycles C ; a cycle is *unbalanced* if its sign is -1 . There are 2^{m-n+1} switching classes. Write $\rho(A_\sigma) = \max_i |\lambda_i(A_\sigma)|$. For a d -regular graph, the conjecture of Bilu and Linial [1] asserts that $\min_\sigma \rho(A_\sigma) \leq 2\sqrt{d-1}$; it is known one-sidedly, hence for bipartite graphs [4], and two-sidedly with the constant $2\sqrt{3(d-1)}$ [5].

The circulant $C_n(1, 2)$ has vertex set $\mathbb{Z}_n$ and edges $\{i, i+1\}$ (step 1) and $\{i, i+2\}$ (step 2); it is 4-regular, with Kesten bound $2\sqrt{3} \approx 3.464$. Its quadrilaterals, for $n \geq 10$, are exactly $Q_i = (i, i+1, i+3, i+2)$, one for each $i \in \mathbb{Z}_n$, and its triangles are $T_i = (i, i+1, i+2)$. At $n = 8$ the step-2 edges close into two further quadrilaterals, $(0, 2, 4, 6)$ and $(1, 3, 5, 7)$; we assume $n \geq 10$ throughout and treat that case in Remark 4. We study the $\mathbb{F}_2$ system

$$s \cdot \chi_{Q_i} = 1 \quad (i \in \mathbb{Z}_n), \tag{1}$$

requiring every quadrilateral to be unbalanced. This is the parity family of the companion manuscript [2] specialized to $C_n(1, 2)$; there, averaging over such families is developed into a general method for the Bilu-Linial conjecture, with short even cycles shown to control the even traces of A_σ at first order. The present graphs furnish its exactly solvable case.

*Companion note to [2]. Verification suite, drivers, and experimental data: <https://github.com/Vaibhavs25/bilu-linial-parity>

2 Results

Proposition 1. *Let $n \geq 10$ be even. On $C_n(1, 2)$ the system (1) is consistent and its solutions form exactly four switching classes. The class containing the signing $\sigma \equiv +1$ on step-1 edges $\{i, i+1\}$ and $\sigma_{\{i, i+2\}} = (-1)^i$ on step-2 edges has spectrum*

$$\left\{ \pm 2\sqrt{\cos^2 \theta_k + \cos^2 2\theta_k} : \theta_k = \frac{2\pi k}{n}, 0 \leq k < \frac{n}{2} \right\},$$

so its spectral radius equals $2\sqrt{2}$ exactly, against the Kesten floor $2\sqrt{3}$. For odd n the system is inconsistent, and at most $n - 1$ quadrilaterals can be simultaneously unbalanced.

Proof. The quadrilaterals of $C_n(1, 2)$ are the Q_i , one per vertex, with sign $a_i a_{i+2} b_i b_{i+1}$, where a_i, b_i denote the signs of $\{i, i+1\}$ and $\{i, i+2\}$. The displayed signing gives $a_i a_{i+2} b_i b_{i+1} = (-1)^i (-1)^{i+1} = -1$ for all i , so the system is consistent. If $\sum_{i \in S} \chi_{Q_i} = 0$, then comparing coefficients of the step-2 edge $\{i, i+2\} \in Q_i \cap Q_{i-1}$ forces S to be shift-invariant, so $S = \emptyset$ or $S = \mathbb{Z}_n$; the full sum is indeed zero (each edge lies in exactly two quadrilaterals). Hence the constraint map on the $2^{m-n+1} = 2^{n+1}$ switching classes has rank $n - 1$, and the solution set is a coset of dimension 2: four classes.

For the displayed class, write C_1, C_2 for the unsigned step-1 and step-2 circulants and $D = \text{diag}((-1)^j)$; then $A_\sigma = C_1 + DC_2$, which is symmetric since D commutes with the even shift C_2 . On the orthonormal Fourier basis $f_k(j) = n^{-1/2} e^{2\pi i j k / n}$ we have $C_1 f_k = 2 \cos \theta_k f_k$, $C_2 f_k = 2 \cos 2\theta_k f_k$, and $D f_k = f_{k+n/2}$. Each plane $\text{span}\{f_k, f_{k+n/2}\}$ is therefore invariant, carrying the block

$$\begin{pmatrix} 2 \cos \theta_k & 2 \cos 2\theta_k \\ 2 \cos 2\theta_k & -2 \cos \theta_k \end{pmatrix},$$

where we used $\cos \theta_{k+n/2} = -\cos \theta_k$ and $\cos 2\theta_{k+n/2} = \cos 2\theta_k$. A traceless symmetric 2×2 block $\begin{pmatrix} a & b \\ b & -a \end{pmatrix}$ has eigenvalues $\pm \sqrt{a^2 + b^2}$, giving the stated spectrum. Setting $u = \cos^2 \theta \in [0, 1]$, $\cos^2 \theta + \cos^2 2\theta = 4u^2 - 3u + 1 \leq 2$ with equality iff $u = 1$, attained at $k = 0$; hence $\rho = 2\sqrt{2}$. For odd n , summing all n constraints gives 0 on the left (each edge appears twice) and $n \equiv 1 \pmod{2}$ on the right. $\square$

Proposition 2 (Twisted classes). *Let τ_i denote the sign of the triangle $T_i = (i, i+1, i+2)$ and α the sign product around the step-1 Hamilton cycle H . Then $\chi_{Q_i} = \chi_{T_i} + \chi_{T_{i+1}}$, so (1) is equivalent to the alternation $\tau_{i+1} = -\tau_i$; the triangles together with H form a basis of the cycle space, so the four solution classes are coordinatized by $(\tau_0, \alpha) \in \{\pm 1\}^2$; and rotation by one vertex is an automorphism carrying (τ_0, α) to $(-\tau_0, \alpha)$. Consequently ρ depends on α alone: it equals $2\sqrt{2}$ for $\alpha = +1$, and for $\alpha = -1$ it equals*

$$\rho_-(n) := 2\sqrt{\cos^2(\pi/n) + \cos^2(2\pi/n)} < 2\sqrt{2}.$$

In particular $\min_\sigma \rho(A_\sigma) \leq \rho_-(n)$ for every even n .

Proof. Each step-2 edge b_i lies in T_i only, so $\sum_{i \in S} \chi_{T_i} = 0$ forces $S = \emptyset$; together with the fact that no combination of triangles has empty step-2 support except the empty one, $\{\chi_{T_0}, \dots, \chi_{T_{n-1}}, \chi_H\}$ is a basis of the $(n+1)$ -dimensional cycle space, giving the coordinates; the rotation statement is immediate since rotation shifts the alternating pattern by one. For $\alpha = -1$ it therefore suffices to treat the representative with $\tau_i = (-1)^i$. Let R be the cyclic shift, $D = \text{diag}((-1)^j)$, $\varphi = \pi/n$, and set

$$A(\varphi) = e^{i\varphi} R + e^{-i\varphi} R^* + e^{2i\varphi} D R^2 + e^{-2i\varphi} R^{-2} D,$$

a Hermitian matrix supported on $C_n(1,2)$. Its holonomy on the directed triangle $i \rightarrow i+1 \rightarrow i+2 \rightarrow i$ is $e^{i\varphi} \cdot e^{i\varphi} \cdot (-1)^i e^{-2i\varphi} = (-1)^i$, and on H it is $e^{in\varphi} = -1$; two Hermitian matrices with common connected support, all nonzero entries of modulus one, and equal holonomy on every cycle are conjugate by a diagonal unitary (both conditions hold here), so $A(\varphi)$ is isospectral with the real signed adjacency matrix of the class $(+1, -1)$. Note the step-2 phase is forced to be 2φ : this is exactly the co-shift that keeps the triangle holonomies real, and it is the step omitted by a naive one-parameter Bloch argument. As in Proposition 1, the planes $\text{span}\{f_k, f_{k+n/2}\}$ are invariant, with blocks

$$\begin{pmatrix} 2\cos(\theta_k + \varphi) & 2\cos(2\theta_k + 2\varphi) \\ 2\cos(2\theta_k + 2\varphi) & -2\cos(\theta_k + \varphi) \end{pmatrix},$$

hence spectrum $\pm 2\sqrt{g(\theta_k + \pi/n)}$ with $g(\theta) = \cos^2 \theta + \cos^2 2\theta$ and shifted momenta $\theta_k + \pi/n \in \{(2k+1)\pi/n\}$, which avoid 0 and π . Writing $g = 4u^2 - 3u + 1$ with $u = \cos^2 \theta$, g decreases on $[0, \theta_0]$ and increases on $[\theta_0, \pi/2]$, where $\cos 2\theta_0 = -\frac{1}{4}$, with $g(\pi/2) = 1$ and g symmetric under $\theta \mapsto \pi - \theta$; since $g(\pi/n) \geq g(\pi/8) = \frac{4+\sqrt{2}}{4} > 1$ for every even $n \geq 8$, the maximum over the shifted lattice is attained at π/n , giving $\rho_-(n)$. (We state the inequality from $n = 8$ so that Remark 4 may invoke it.) $\square$

Conjecture 3. *For every even $n \geq 8$, $\min_{\sigma} \rho(A_{\sigma}) = \rho_-(n)$ on $C_n(1,2)$: the $\alpha = -1$ twisted class is the global optimizer. This is verified by exhaustive enumeration of all 2^{n+1} switching classes for $n \in \{8, 10, 12, 14, 16, 18\}$, with agreement to 10^{-9} . The lower bound over all classes is a flux-minimization statement in the spirit of Lieb’s flux-phase theorem [3]: among all $\mathbb{F}_2$ flux assignments to the triangle-Hamilton cycle basis, alternating triangle flux with anti-periodic step-1 holonomy minimizes the spectral radius. On this family the quadrilateral rule thus contains the (conjecturally exact) global optimizer, while random signings concentrate above the Kesten floor $2\sqrt{3}$ from $n \approx 480$.*

Remark 4 (The exceptional case $n = 8$). Identifying (1) with “every quadrilateral of $C_n(1,2)$ is unbalanced” - and hence with the parity family of [2] - requires $n \geq 10$. At $n = 8$ the step-2 edges form two additional quadrilaterals, $S_0 = (0, 2, 4, 6)$ and $S_1 = (1, 3, 5, 7)$, which the canonical signing of Proposition 1 balances. All statements above remain true at $n = 8$ for the system (1) exactly as written: the proofs of Propositions 1 and 2 apply verbatim (rank $7 = n - 1$, four classes, $\rho \in \{2\sqrt{2}, \rho_-(8)\}$, the lattice maximum covered by $g(\pi/8) > 1$). The strictly larger system requiring all ten quadrilaterals to be unbalanced is resolved by a flux computation. Over $\mathbb{F}_2$,

$$\chi_{S_0} = \chi_H + \sum_{i \text{ even}} \chi_{T_i}, \quad \chi_{S_1} = \chi_H + \sum_{i \text{ odd}} \chi_{T_i},$$

since each sum on the right covers every step-1 edge exactly once and the four step-2 edges of the named quadrilateral exactly once. On the solution family $\tau_i = \tau_0(-1)^i$, so $\prod_{i \text{ even}} \tau_i = \prod_{i \text{ odd}} \tau_i = +1$ and both extra quadrilaterals carry sign α : the two additional constraints are jointly equivalent to $\alpha = -1$. The all-ten system therefore has exactly the two twisted classes $(\pm 1, -1)$ as its solutions, both attaining

$$\rho_-(8) = 2\sqrt{\cos^2(\pi/8) + \cos^2(\pi/4)} = \sqrt{4 + \sqrt{2}} = 2.326846\dots,$$

and it excludes the $2\sqrt{2}$ class. Conjecture 3 is unaffected and was additionally verified at $n = 8$ along with the other sizes.

Remark 5. The spectral radius is constant on each of the four classes of the solution family, so sampling the family sees at most the two values $\{\rho_-(n), 2\sqrt{2}\}$ of Proposition 2. By contrast, uniformly random signings of $C_n(1, 2)$ concentrate *above* the Kesten bound for $n \gtrsim 480$, while the family sits at $2\sqrt{2}$ and below, an instance of the general phenomenon that uniform averaging cannot certify a sub-Kesten signing whereas the parity family can [2]. Conjecture 3 asks that alternating triangle flux with anti-periodic step-1 holonomy minimize the spectral radius over all 2^{n+1} flux assignments, the discrete analogue, for signed circulants, of Lieb’s flux-phase theorem [3]. Reflection positivity on $C_n(1, 2)$ is the natural route to the lower bound and is left open.

References

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